

CSE 318 Assignment-04

Decision Tree

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A Decision Tree helps us to make decisions by mapping out different choices and their possible outcomes. It's used in machine learning for tasks like classification and prediction.

There are a number of strategies that we can follow while choosing the attribute in decision tree to explore . In this assignment we used INFORMATION GAIN , INFORMATION GAIN RATIO , NORMALIZED WEIGHTED INFORMATION GAIN .

1 CODE BASE

I used JAVA as my source code .

2 GRAPHS

To evaluate and compare the performance of different attribute selection criteria, we plotted graphs based on experimental results. Each criterion—Information Gain (IG), Information Gain Ratio (IGR), and Normalized Weighted Information Gain (NWIG)—was tested across various maximum depths for two datasets: `Iris.csv` and `adult.data`.

The first set of graphs shows how the average classification accuracy changes with tree depth. The second set illustrates how the number of nodes grows with depth, which reflects the complexity of the learned tree structure.

IRIS.CSV
AVERAGE ACCURACY VS MAX TREE DEPTH

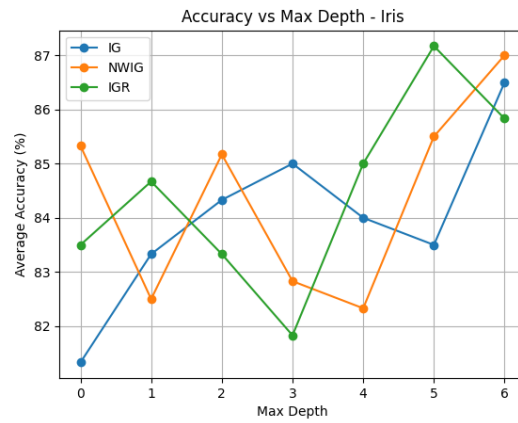


Figure 1: Caption

NUMBER OF NODES VS DEPTH

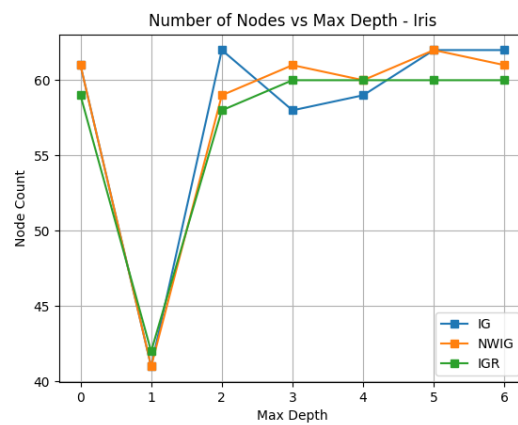


Figure 2: Caption

ADULT.DATA
AVERAGE ACCURACY VS MAX TREE DEPTH

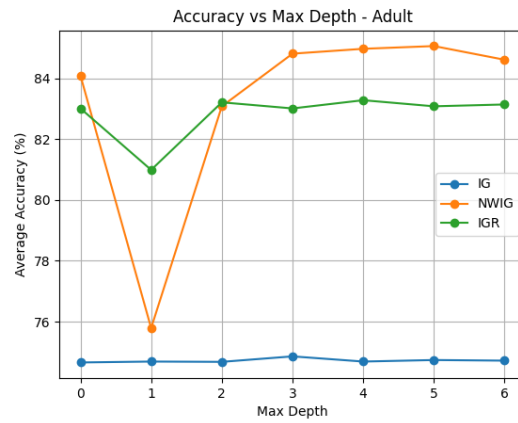


Figure 3: Caption

NUMBER OF NODES VS DEPTH

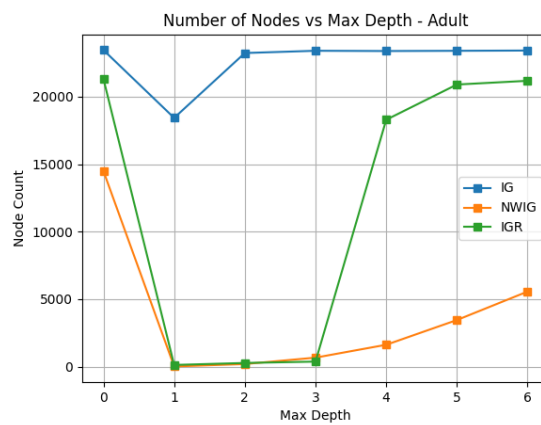


Figure 4: Caption

3 COMPARISON

DEPTH	IRIS			ADULT		
	BEST PERFORMER CRITERION	ACCURACY	NODE COUNT	BEST PERFORMER CRITERION	ACCURACY	NODE COUNT
0	NWIG	85.33%	61	NWIG	84.09%	14483
1	IGR	84.67%	42	IGR	80.99%	118
2	NWIG	85.17%	59	IGR	83.21%	265
3	IG	85%	58	NWIG	84.81%	661
4	IGR	85%	60	NWIG	84.97%	1614
5	IGR	87.17%	60	NWIG	85.06%	3447
6	NWIG	87%	61	NWIG	84.61%	5557

Table 1: Comparison of Best Criterion, Accuracy, and Node Count for Different Depths

4 OBSERVATION

From the comparison table, several patterns emerge regarding the performance of different attribute selection criteria across varying tree depths.

- For both datasets, NWIG often outperforms the others, especially at depth 0, where the tree is unpruned. This suggests that NWIG is well-suited for deeper trees where more information is available to guide splits.
- IGR consistently performs well on the *Iris* dataset, especially at moderate depths like 1, 4, and 5. This may be due to IGR’s ability to handle features with many distinct values, which is common in that dataset.
- IG shows competitive accuracy at lower depths, particularly at depth 3 for *Iris*, but becomes less dominant in the *Adult* dataset, where NWIG overtakes it in deeper trees.
- Pruning (by setting a small max depth) reduces the number of nodes significantly and often helps prevent overfitting. For example, at depth 1, node counts drop drastically while accuracy stays relatively high.
- A clear trade-off is visible between model complexity (measured by node count) and accuracy. Deeper trees tend to have better accuracy but come at the cost of significantly larger size, especially in the *Adult* dataset.
- **Overall, NWIG is the most stable and consistently high-performing criterion across both datasets and various depths. IGR is a good alternative .**

5 CONCLUSION

In this assignment, we implemented and analyzed a decision tree classifier using three different attribute selection strategies: Information Gain (IG), Information Gain Ratio (IGR), and Normalized Weighted Information Gain (NWIG). Through experiments on two datasets—*Iris.csv* and *adult.data*—we observed how each criterion affects classification accuracy, tree complexity, and overall generalization.

Our results show that NWIG performs consistently well across various tree depths, especially in deeper, unpruned trees. IGR, on the other hand, offers a strong balance between accuracy and overfitting control, particularly on the *Iris* dataset. IG occasionally performs competitively but is generally outperformed by the other two in complex datasets like *Adult*.

By plotting accuracy and node count against depth, we were able to clearly visualize the trade-offs between model simplicity and predictive performance. Pruning the tree by limiting depth helps reduce overfitting and model size, while deeper trees offer marginal accuracy gains at the cost of complexity.

Overall, this experiment reinforces the value of using advanced splitting criteria and tuning tree depth to achieve robust and efficient decision tree models.